

Phase I: Project Brainstorming. Due **Sep 22, 2026.**

- Set up a shared workspace for your project files and give your mentors (the PhD TA and undergrad CA in your section) access to it. You may use a Git repository (for example, GitHub), Deepnote, or Google Colab, as long as course staff can access your project. Include a link to it in your submission. If your workspace is private, you will need to add your mentors before the deadline (we will send instructions for how to do this). If your mentors are not given access prior to the deadline, your group will not receive credit for *Phase I*.
- Meet with your group. Write down at least three ideas for datasets. Include as much information as possible about the availability of data. Keep the data sourcing requirement in mind as you brainstorm: the dataset you turn in for the final project must be created through API access, web scraping, or a combination of at least two meaningfully distinct public datasets.
- As a team, **submit a pdf file** on Gradescope that contains a link to your **project workspace** and your **dataset ideas**. Include a “**Questions for reviewers**” section at the end of your submission, listing specific questions for your project mentors to answer in giving you feedback on this phase.
- Ensure that your Gradescope submission includes all members of the project group (so, each project group member should see the submission in their own Gradescope portals).

Submission checklist: To receive credit for *Phase I*, the submission must include the following:

- A link to your group’s project workspace (a Git repository, Deepnote, or Google Colab), with your mentors given access
- Three ideas for datasets. For each idea, include:
 - A short description of the dataset.
 - A short discussion of the *availability* of the data
 - You **must** include links to online data sources, and each idea must be sourceable through API access, web scraping, or a combination of at least two meaningfully distinct public datasets. It is acceptable to include:
 - Link(s) to already formatted data (i.e., data you can download directly).
 - A single downloadable dataset does not meet the requirement on its own. Link at least two meaningfully distinct public datasets and briefly describe how you would combine them: what shared unit of observation or key you would join on (e.g. county, player-season, date), and what the combined data lets you ask that neither source answers alone.
 - Link(s) to websites from which you intend to collect data. If you intend to collect data, you must also include a brief description of *how* you plan to collect the data. For example:
 - Will you use an API? If so, please make sure to discuss whether you need an API key and whether the API will cost money to use (we recommend that you use free APIs)
 - Will you scrape the data? If so, please make sure to confirm whether the website allows web scraping.
 - A short discussion of the *trustworthiness* of the data. This might include:
 - Who collected the data? Do you trust this source? Why? If you plan to combine two or more datasets, answer this for each source.
 - If the dataset you are using was published (for example, in an academic paper or a government report), make sure to cite that publication as part of your discussion.

- How were the data checked for quality? Do you have any concerns about the accuracy of the data?
- Questions for reviewers

The following Phase 1 rubric includes recommendations for ways to start the final project with a strong foundation. We recommend using this for self-evaluation.

Category	Least developed projects <i>0 points</i>	Less developed projects <i>1 point</i>	Typical projects <i>2 points</i>	More developed projects <i>3 points</i>
Project workspace	A link to the project workspace is not included	A link is included but not all group members / mentors have access.	A link is included, all members of the group have access, and an invite has been sent to the mentors.	A link is included and all members of the group have practiced working in the shared workspace (for example, pushing and pulling from a repository). The mentors have access. The group has created folders or a readme to stay organized.

Dataset ideas <i>(graded for each idea)</i>	Fewer than three dataset ideas are included.	Three datasets ideas are included, but dataset ideas are vague and impossible or excessively difficult to collect. Some datasets are not described or do not have notes about availability with a source cited. It is unclear whether the ideas meet the data sourcing requirement (API access, web scraping, or a combination of at least two meaningfully distinct public datasets).	Three datasets ideas are included and all or most datasets could feasibly be collected or accessed by the end of the semester. Each dataset is described alongside a note about availability with a source cited. All or most ideas meet the data sourcing requirement.	Three datasets ideas are included and all datasets could feasibly be collected or accessed by the end of the semester. Each dataset is described alongside a note about availability with (possibly multiple) sources cited. Each dataset could reasonably be part of a data science project, driven by an interesting research question. Each idea clearly meets the data sourcing requirement, and the collection method (API or web scraping) or the plan for combining at least two meaningfully distinct public datasets is described concretely.
Questions for reviewers	There are no questions for reviewers included.	The questions for reviewers are vague or unclear.	The questions for reviewers are specific to the datasets and are based on group discussions between team members.	The questions for reviewers are specific to the datasets and are based on group discussions between team members. Questions for reviewers look toward the next stage of the project (<i>Phase II</i>).